

Paper name	Year	Dataset & Size	Language & Script	Platform	Task Family	ML Methods	DL & Transformer Models	Cross Domain	Metrics & Scores	Main Finding / Result
Idiots at HASOC 2019: Fine-tuning Transformer Neural Networks for Hate Speech Identification in Indo-European Languages	2019	HASOC 2019 (DE: Train/Dev 3819, Test 850; EN: Train/Dev 5852, Test 1153; HI: Train/Dev 4665, Test 1318)	English; German; Hindi [Not Reported]	Twitter; Facebook	Hate Speech & Offensive Content Identification [Hate Type: Both]	None	DL: Transformer Trans: BERT; bert-base-german-cased; bert-base-multilingual-cased; bert-base-cased; bert-base-uncased; bert-base-multilingual-uncased	No	Metrics: macro-f1; weighted f1 Macro-F1: Yes	Won first place for English sub-tasks A and B, and Hindi sub-task B; within top 5 for 7 out of 8 tasks.
A Comparative Study of Different State-of-the-Art Hate Speech Detection Methods in Hindi-English Code-Mixed Language	2020	DATA SET-1 (Bohra 2018); DATA SET-2 (HASOC); DATA SET-3 (ours) (SET-1: 4579; SET-2: 4665; SET-3: 3367)	Hinglish-Code-Mixed [Both (Devanagari & Roman)]	Twitter; Facebook	Hate Speech Detection [Hate Type: Both]	SVM; MNB; KNN; DT	DL: Character-label-CNN	Yes	Metrics: Accuracy; Micro F1 score Acc: Yes	Character-level CNN model performs best across all three combined achieved 86% accuracy and 0.74 Micro F1
A dataset of hindi-english code-mixed social media text for hate speech detection.	2018	Hinglish Hate (4575 tweets)	Hinglish-Code-Mixed [Roman]	Twitter	Hate Speech Detection [Hate Type: Both]	Random Forest Classifier	None	No	Metrics: Accuracy Acc: Yes	Best accuracy of 69.9% achieved using Word N-Grams and a Random Forest Classifier.
A framework for online hate speech detection on code-mixed hindi-english text and hindi text in devanagari.	2023	Transliterated Hinglish; Annotated Devanagari Dataset (Hinglish: 11,835 tweets; Devanagari: 4,665 posts)	Hinglish-Code-Mixed; Hindi [Both]	Twitter; Facebook	Hate Speech Detection [Hate Type: Both]	SVM; Decision Trees; Boosting; Random Forest; XGBoost	DL: DNN; LSTM; TabNet Trans: MuRIL; XLM-RoBERTa; mBERT; Indic-BERT	Yes	Metrics: Accuracy; Precision; Recall; F1-Score Macro-F1: Yes Acc: Yes	TabNet + PCA + MuRIL achieved the best accuracy of 90% on Hinglish data and 85% on Devanagari text.
A multilingual, multimodal dataset of aggression and bias: the ComMA dataset	2024	ComMA Dataset v0.2 (57,363 comments, 1,142 memes, 70h audio)	Meitei; Bangla; Hindi; Indian English [Both; Other]	YouTube; Facebook; Twitter; Telegram	Aggression and Bias Identification [Hate Type: Both]	None	DL: Transformer Trans: Indic-BERT; mBERT; DistilBERT; XLM-RoBERTa; MuRIL	Yes	Metrics: macro avg. f1-score Macro-F1: Yes	XLM-RoBERTa-base achieved best F1-scores of 0.58 for aggression and 0.52 for gender bias.
A transformer-based multi-task framework for joint detection of aggression and hate on social media data	2023	HASOC-2019; TRAC-2 2020 (HASOC EN (5852/1153); HASOC HI (4665/1318); TRAC-2 HI (5000/1200))	English; Hindi; German; Bengali [Both; Other]	Twitter; Facebook; YouTube	Aggression, Hate, and Offense Identification [Hate Type: Both]	None	DL: Transformer Trans: XLM-RoBERTa; mBERT; BERT	Yes	Metrics: macro-F1; weighted F1; Accuracy Macro-F1: Yes Acc: Yes	Proposed multi-task MTFHAD system outperforms single-task baselines across multiple dimensions.
An ensemble model for hate speech and offensive content identification in indo-european languages.	2020	HASOC 2020 (EN (3708/814); DE (2373/526); HI (2963/663))	English; German; Hindi [Not Reported]	Twitter; Facebook	Hate and Offensive Content Identification [Hate Type: Both]	Random Forest; Gradient Boosting; XGBoost	None	No	Metrics: F1 Macro average Macro-F1: Yes	Soft voting ensemble of RF, GB, and XGBoost achieved competitive results across both subtasks.
An Indian language social media collection for hate and offensive speech.	2020	PEI 2019; SemEval 2019 Task 6; HASOC (PEI 2019: ~2000 posts (Train 1519, Test 488))	English; Hindi; Hinglish-Code-Mixed [Not Reported]	Twitter; Facebook	Hate Speech and Offensive Content Identification [Hate Type: Both]	MNB; SGD; LR; Linear SVM	None	Yes	Metrics: Precision; Recall; F1-score; Accuracy Macro-F1: Yes Acc: Yes	SGD classifier achieved the best accuracy (PEI English Sub-task A: 0.76 Macro F1, 0.78 accuracy).

Automatic detection of hate speech in code-mixed indian languages in twitter social media interaction using DConvBLSTM-MuRIL ensemble method.	2024	CBLSTM-Code-Mixed; HEOT; FIRE2021 (Hinglish: 6,370; Tamil-EN: 7,401; Malayalam-EN: 7,320)	Hinglish-Code-Mixed; Other [Roman]	Twitter	Hate Speech and Offensive Language Detection [Hate Type: Both]	LR; RF; MNB; XGB; SVM	DL: CNN; LSTM; Bi-LSTM; CNN-LSTM; DConvBLSTM Trans: BERT; mBERT; DistilBERT; XLM-RoBERTa; IndicBERT; MuRIL	Yes	Metrics: F1-scores; Accuracy; RMSE; MAPE; MAE Macro-F1: Yes Acc: Yes	DConvBLSTM + MuRIL ensemble outperformed all models (Hinglish Hate F1: 0.948, Tamil-EN F1: 0.943).
Automatic hate speech detection using ensemble method and natural language processing techniques.	2023	Kaggle English; GitHub Hinglish (Not Reported)	English; Hinglish-Code-Mixed [Roman]	Twitter	Hate, Offensive or Neither Prediction [Hate Type: Both]	Random Forest; SVC; MNB; KNN; Logistic Regression	None	Yes	Metrics: Accuracy Acc: Yes	Ensemble of RF and SVC using TF-IDF Vectorizer achieved accuracy of 90.7%.
CHUNAV: Analyzing hindi hate speech and targeted groups in indian election discourse.	2024	CHUNAV (11,457 Hindi tweets)	Hindi [Devanagari]	Twitter	Hate Speech Detection and Target Identification [Hate Type: Both]	Naive Bayes; Decision Tree; SVM; Logistic Regression	DL: Conv-BiLSTM; BiLSTM Trans: XLM-RoBERTa; BERT (HAM); BERT (HCMAM); RoBERTa-Hasoc	No	Metrics: Accuracy; MMAE; F1-Score Macro-F1: Yes Acc: Yes	Hard Ensemble of BERTs (HEB) achieved the best F1 of 0.726 (Accuracy: 0.765) for Target Identification.
CMHE-AN: Code mixed hybrid embedding based attention network for aggression identification in hindi english code-mixed text.	2023	TRAC 2-2020 Hi-En; Hi-En code mixed dataset (Dataset 1: 5184 posts; Dataset 2: 3659 posts)	Hinglish-Code-Mixed [Roman]	YouTube; Twitter	Aggression Identification; Hate Speech Detection [Hate Type: Both]	Logistic Regression; XGBoost; Naive Bayes	DL: char CNN; word CNN; subword LSTM; CMHE-AN Trans: mBERT	Yes	Metrics: precision; recall; accuracy; f1-score (weighted avg) Acc: Yes	Proposed BiLSTM-Attention (CMHE-AN) model achieved best accuracy 77.54% (TRAC-2), 735.23% (Hinglish) .
Ceasing hate with moh: Hate speech detection in hindi-english code-switched language.	2022	TRAC-I; HOT; HS (TRAC-I: 11,623; HOT: 2,803; HS: 4,575)	Hinglish-Code-Mixed [Both]	Twitter; Facebook	Hate Speech and Offense Detection [Hate Type: Both]	Naive Bayes; LR; SVM; RF; XGBoost; AdaBoost; Gradient Boosting	DL: CNN-1D; Sub-Word Level LSTM; Hierarchical LSTM Trans: mBERT; MuRIL	Yes	Metrics: F1 score; Precision; Recall; Accuracy Macro-F1: Yes Acc: Yes	MoH + MuRIL outperformed all baselines on TRAC-I (F1: 0.71), HS (F1: 0.85), and HOT (F1: 0.90).
Code-switching patterns can be an effective route to improve performance of downstream NLP applications: A case study of humour, sarcasm and hate speech detection.	2020	Humour; Sarcasm; Hate (Humour: 3,453; Sarcasm: 5,250; Hate: 4,575)	Hinglish-Code-Mixed [Roman]	Twitter	Sentiment, Humour, Sarcasm, and Hate Speech Detection [Hate Type: Both]	Not Reported	DL: Not Reported	No	Metrics: Not Reported	Demonstrates that code-switching patterns serve as an effective feature for downstream NLP.
Code-mixed sentiment and hate-speech prediction.	2024	FrenchBookReviews; FrenchOLiD; TamilCMSenti; TamilCMOLiD; IliTH; Hinglish Hate; etc. (TamilCMSenti: 36,881; TamilCMOLiD: 36,881; IliTH: 36,881)	Hinglish-Code-Mixed; Other [Both; Other]	Twitter; YouTube; Other	Sentiment Analysis; Offensive Language Detection [Hate Type: Both]	None	DL: Transformer Trans: mBERTc-base; XLM-R-base; TWHIN-BERT; SIEng-BERT; SloBERTa-SIEng; MuRIL-en-hi-codemixed; HingRoBERTa	Yes	Metrics: Macro F1 score Macro-F1: Yes	Bilingual/multilingual transformer models specialized for social media outperformed monolingual and huge generative models.
Comparative study of pre-trained bert models for code-mixed hindi-english data.	2023	Icon; Emotions; Sentiment; Hatespeech; HASOC (Icon: 18,461; Emotions: 151,311; Sentiment: 3,879;	Hinglish-Code-Mixed [Roman]	Twitter; social media	Sentiment, Emotion, and Hate Speech Identification [Hate Type: Both]	None	DL: Transformer Trans: HingBERT; HingRoBERTa; HingRoBERTa-Mixed; mBERT; ALBERT; BERT; RoBERTa	Yes	Metrics: Accuracy; Precision; Recall; F1 score (Macro) Macro-F1: Yes Acc: Yes	HingRoBERTa-based models specifically trained on code-mixed text achieved top F1-scores across all tasks.

Comparison of pretrained embeddings to identify hate speech in indian code-mixed text.	2020	Hinglish Hate Speech Dataset (4,575 sentences)	Hinglish-Code-Mixed [Not Reported]	Twitter	Hate Speech Detection [Hate Type: Both]	Hard Voting	DL: CNN; Transformer Trans: BERT; RoBERTa; XLNet; DistilBERT	No	Metrics: Precision; Recall; F1 Score; Macro average; Weighted average; Accuracy Macro-F1: Yes Acc: Yes	XLNet achieved the best overall performance (Macro F1 of 0.67, Accuracy of 0.73).
Faux hate: Unravelling the web of fake narratives in spreading hateful stories: A multi-label and multi-class dataset in cross-lingual hindi-english code-mixed text.	2024	Faux Hate Multi-Label Data set (FHMLD) (8,014 comments)	Hinglish-Code-Mixed [Both]	Twitter; YouTube	Fake-Instigated Hate Speech Detection [Hate Type: Both]	TF-IDF + SVM; word2vec + SVM; fastText + SVM	DL: CNN; LSTM Trans: BERT + SVM; mBERT + SVM; HateBERT + SVM	No	Metrics: Accuracy; Macro F1 score; ROC; AUC Macro-F1: Yes Acc: Yes	fastText + SVM outperformed transformer models (Accuracy: 81% for fake, 82% for hate prediction).
HHSD: Hindi hate speech detection leveraging multi-task learning.	2023	Hindi Hate Speech Dataset (HHSD) (Approximately 15K tweets)	Hindi [Devanagari]	Twitter	Hindi Hate Speech Detection [Hate Type: Both]	None	DL: CNN; Bi-LSTM; MTL (Multi-Task Learning) Trans: mBERT; LaBSE; MuRIL; XLM-RoBERTa	Yes	Metrics: Accuracy; Weighted-F1; Exact match; Hamming loss Acc: Yes	Multi-task learning framework with Combined Hindi-Bangla-Urdu (HBU) data achieved 82.6% accuracy.
DLRG@ HASOC 2019: An enhanced ensemble classifier for hate and offensive content identification.	2019	HASOC 2019 (DE: 3819 Train / 850 Test; HI: 4665 Train / 1318 Test)	German; Hindi [Not Reported]	Twitter	Hate and Offensive Content Identification [Hate Type: Both]	Logistic Regression; Support Vector Machine; Random Forest Classifier	None	No	Metrics: Accuracy; CHI square; Mutual Information Acc: Yes	Ensemble Random Forest classifier with CHI square based feature selection achieved 81% on German and 64% on Hindi.
Detecting offensive tweets in Hindi-English code-switched language.	2018	(HETO); Dataset A (HOT: 3,679; Dataset A: 14,509)	Hinglish-Code-Mixed; English [Roman]	Twitter	Offensive Tweet Detection [Hate Type: Both]	None	DL: CNN	Yes	Metrics: Accuracy Acc: Yes	Ternary Trans-CNN (transfer learning) achieved 83.9% accuracy, baseline 58.7%
HateCheckHIn: Evaluating Hindi Hate Speech Detection Models.	2022	HateCheckHIn (14,318 tweets combined (4,468 Hate, 1,416 Non-Hate))	Hindi; Hinglish-Code-Mixed [Both]	Twitter	Hate Speech Detection [Hate Type: Both]	SVM; LR; NB; KNN	None	Yes	Metrics: Accuracy; Precision; Recall; F1-Score Macro-F1: Yes Acc: Yes	SVM with TF-IDF Vectorization achieved accuracy of 0.96 (Hate F1: 0.99, Non-Hate F1: 0.99)
Detection of hate speech text in hindi-english code-mixed data.	2020	Hinglish Code-mixed Dataset (10,000 texts)	Hinglish-Code-Mixed [Both]	Twitter; Facebook	Hate Speech Detection [Hate Type: Both]	SVM; Random Forest	DL: fastText	Yes	SVM-RBF with fastText achieved 85.8% accuracy	Sentence vector representations were created using the mean of fastText word embeddings.
Detection and classification of aggressive comments and hate speech.	2023	Cyberbullying 2020 dataset; Devanagari Hindi; Hinglish (English: 47k comments)	English; Hinglish-Code-Mixed; Hindi [Both]	Twitter	Cyberbullying Detection [Hate Type: Both]	None	DL: CNN; Bi-LSTM Trans: BERT; BERT-base (uncased)	Yes	Metrics: Accuracy Acc: Yes	The BERT model achieved approximately 10% higher accuracy compared to the CNN model.
Did you offend me? Classification of offensive tweets in hinglish language.	2018	Hinglish Offensive Tweets (HOT) (3189 tweets)	Hinglish-Code-Mixed [Roman]	Twitter	Offensive Language Classification [Hate Type: Both]	SVM; Random Forests	DL: CNN; LSTM; MIMCT (parallel CNN-LSTM)	Yes	Metrics: Precision; Recall; F1 Macro-F1: Yes Acc: Yes	MIMCT model achieved peak F1 of 0.895 using concatenated embeddings and metadata
Does aggression lead to hate? Detecting and reasoning offensive traits in hinglish code-mixed texts.	2022	Hinglish code-mixed dataset (consolidated 5 sources) (Aggression: 30,000; Hate: 4,578; Humor: 2,052; Sarcasm: 2,052)	Hinglish-Code-Mixed [Roman]	Twitter; Facebook	Offense Detection (Aggression, Hate, Humor, Sarcasm, Stance) [Hate Type: Both]	LR; RF; SVM; XGB; LGBM	DL: WLSTM; W-HAN; MMNSS; CNN-LSTM; CS-ELMO; C-HAN; Transformer Trans: BERT; Transformer (custom trained on	Yes	Metrics: Weighted-F1; Macro-F1; Confusion Matrices Macro-F1: Yes Acc: Yes	Proposed custom Transformer and C-HAN outperformed multilingual pre-trained BERT variants.

Hate speech detection from code-mixed hindi-english tweets using deep learning models.	2018	Hinglish Code-Mixed Dataset (Bohra 2018) (4,575 tweets)	Hinglish-Code-Mixed [Roman]	Twitter	Hate Speech Detection [Hate Type: Both]	None	DL: CNN-1D	No	Metrics: F1-score; Accuracy Macro-F1: Yes Acc: Yes	CNN-1D model achieved the highest F1-score of 0.808 (Accuracy of 0.826) on the HS dataset.
Hate speech detection in hindi-english code-mixed social media text.	2019	Hindi-English Code-Mixed Social Media Text (Bohra 2018) (4,575 tweets)	Hinglish-Code-Mixed [Roman]	Twitter	Hate Speech Detection [Hate Type: Both]	SVM Classifier; Random Forest Classifier	DL: sub-word level LSTM; Hierarchical LSTM with attention	No	Metrics: Accuracy; Recall; F1-score Macro-F1: Yes Acc: Yes	Hierarchical LSTM with attention based on phonemic sub-words achieved the best F1 of 48.7%.
Hate speech detection in limited data contexts using synthetic data generation.	2024	English, Hindi, and Vietnamese hate speech datasets (English: 3,000 posts; Hindi: 450 posts limit; Vietnamese: 450)	English; Hindi; Vietnamese [Both; Other]	Twitter; social media	Hate Speech Detection [Hate Type: Both]	None	DL: Transformer Trans: mBERT; BLOOM	Yes	Metrics: Mean macro F1 Macro-F1: Yes	Proposed Contextual Entity Substitution (CES) method outperformed Machine Translation data augmentation.
Hate speech recognition in multilingual text: Hinglish documents.	2023	Consolidated English-Hindi code mix dataset (20,600 instances)	Hinglish-Code-Mixed [Both]	Twitter; YouTube; Facebook	Hate Speech Recognition [Hate Type: Both]	Naïve Bayes; Decision Tree; Logistic Regression; Random Forests; Xgboost; SVM; k-NN	DL: LSTM; BiLSTM; CNN-LSTM; CNN-BiLSTM	Yes	Metrics: Accuracy; Precision; Recall; F1-score; ROC_AUC Macro-F1: Yes Acc: Yes	The proposed CNN-BiLSTM model with word2vec embeddings achieved the best F1 of 0.835
Hatecircle and unsupervised hate speech detection incorporating emotion and contextual semantics.	2023	Hindi HASOC; Hindi DHOT; TRAC2-Hindi-Roman; Bengali FB; etc. (Hindi HASOC: 4,665; Bengali NNTI: 29,999)	Hindi; Hinglish-Code-Mixed; Bengali; Other [Both; Other]	Twitter; Facebook; YouTube; Instagram	Unsupervised Hate Speech Detection [Hate Type: Both]	SVM; Naïve Bayes; Random Forest; Logistic Regression	None	Yes	Metrics: Precision; Recall; F1-Score; Accuracy Macro-F1: Yes Acc: Yes	HateCircle + HateTweet (HC + HT) algorithm outperformed baselines (Hindi TRAC2 F1: 0.74, Bengali NNTI F1: 0.88)
HateDetector: Multilingual technique for the analysis and detection of online hate speech in social networks.	2024	Davidson Dataset; Hinglish Hate Dataset; HOT Dataset (Davidson: 24,783; Hinglish Hate: 4,575; HOT: 3,679)	English; Hinglish-Code-Mixed [Roman]	Twitter	Hate Speech and Offense Detection [Hate Type: Both]	Random Forest; Logistic Regression; AdaBoost	DL: CNN; BERT-CNN; Bi-LSTM; Ternary Trans-CNN; DL Ensemble Stacked; HateDetector Trans: BERT; mBART; GPT4	Yes	Metrics: Accuracy; Precision; Recall; F1-score Macro-F1: Yes Acc: Yes	The proposed HateDetector achieved high performance (97.9% Accuracy, Precision, F1-score, and 98% Recall).
Hate or non-hate: Translation based hate speech identification in code-mixed hinglish data set.	2021	Hindi-English Code-Mixed Social Media Text (Bohra 2018) (4,575 code-mixed tweets)	Hinglish-Code-Mixed [Both]	Twitter	Hate Speech Identification [Hate Type: Both]	SVM; Logistic Regression; Random Forest; Naive Bayes; KNN	DL: TIF-DNN model Trans: mBERT	No	Metrics: Accuracy; F1-hate; F1-nohate Acc: Yes	TIF-DNN model achieved the highest accuracy of 73% and F1-hate of 56%.
Hatred and trolling detection transliteration framework using hierarchical LSTM in code-mixed social media text.	2023	WhatsApp; Twitter; Instagram; Facebook datasets (WhatsApp: 1,259; Twitter: 1,660; Instagram: 1,125; Facebook: 1,861)	Hinglish-Code-Mixed [Roman]	WhatsApp; Twitter; Instagram; Facebook	Hatred and Trolling Detection [Hate Type: Both]	CRF; LR; SVM; Random Forest; Decision Tree	DL: BLSTM; Hierarchical LSTM (HLSTM)	Yes	Metrics: Accuracy; Precision; Recall; F1 Score; Specificity; Sensitivity Macro-F1: Yes Acc: Yes	The proposed HLSTM model achieved the highest accuracy of 97.49% for hatred word detection.
Hindi-english hate speech detection: Author profiling, debiasing, and practical perspectives.	2020	HS (Bohra 2018); HEOT (Mathur 2018b) (HS: 4,575; HOT: 3,679)	Hinglish-Code-Mixed [Both]	Twitter	Hate Speech and Offensive Language Detection [Hate Type: Both]	SVM; Random Forest	DL: seq2seq; CNN; LSTM; BiLSTM; BiLSTM + Attn; CNN + BiLSTM + Attn + PV + DW	Yes	Metrics: Accuracy; F1 score Macro-F1: Yes Acc: Yes	Model combining FastText + CNN + BiLSTM + Attn + PV + DW + Debias achieved top F1 of 0.73 on HS and 0.77 on HOT and 0.78 on HS and 0.85 on HOT

HostileNet: Multilabel hostile post detection in hindi.	2023	CONSTRAINT-2021 shared task dataset (8,192 posts)	Hindi [Devanagari]	Twitter; Facebook	Multilabel Hostile Post Detection [Hate Type: Both]	None	DL: Bi-LSTM; HostileNet Trans: HindiBERT; IndicBERT	Yes	Metrics: F1-scores; Weighted F1-score; Jaccard score; Macro F1-score; Hamming loss	HostileNet outperformed top baselines (Weighted F1: 66.32 on fine-tuning)
IITG-ADBU at HASOC 2019: Automated hate speech and offensive content detection in english and code-mixed hindi text.	2019	HASOC 2019 (EN: 7,005; HI: 5,983 (Train/dev 4665, Test 1318))	English; Hinglish-Code-Mixed [Both]	Twitter; Facebook	Hate Speech and Offensive Content Identification [Hate Type: Both]	Logistic Regression; Support Vector Machine	DL: BiLSTM; Neural Network	No	Metrics: Accuracy; Precision; Recall; Macro F1; Weighted F1 Macro-F1: Yes Acc: Yes	SVM with char n-grams achieved 82.7% accuracy
Improving code-mixed hate detection by native sample mixing: A case study for hindi-english code-mixed scenario.	2025	English-Hindi CM; Hindi (HateCheckHn); English (HASOC-2019) (English-Hindi CM: Train 3211, Validation 687, Test 686 (Total 4,575))	Hinglish-Code-Mixed; Hindi; English [Both]	Twitter	Hate Detection [Hate Type: Both]	SVM; Random Forest; Naive Bayes	DL: Transformer Trans: XLM; mBERT; XLM-R; IndicBERT; MuRIL; trans variants	Yes	Metrics: Accuracy; Precision; Recall; F1-score Macro-F1: Yes Acc: Yes	Adding native samples to the CM training set improved MLM performance (MuRIL_trans achieved F1 of 0.63).
IndicCONAN: A multilingual dataset for combating hate speech in indian context.	2024	IndicCONAN (Over 2,500 examples each for Hindi and Indian English)	English; Hindi [Both; Other]	Twitter; Facebook	Counter-Narrative Generation against Hate Speech [Hate Type: Both]	None	DL: Autoregressive Language Models Trans: Autoregressive language models; XLMR	No	Metrics: Self-BLEU-1; BLEU-2; BLEU-3; ROUGE-1; ROUGE-L; PerspectiveScore; BERTScore; Perplexity	Pipeline (M4) conditioned on HS, HS type, and CN type achieved best performance.
Large scale annotated dataset for code-mix abusive short noisy text. Language Resources and Evaluation	2024	Devanagari-Roman code-mix abusive short noisy text dataset (36,423 code-mix instances)	Hinglish-Code-Mixed [Both]	Twitter	Abusive Language Detection [Hate Type: Both]	Gradient Booster; SGD; LR; KNN; Bernoulli Naive Bayes; Random Forest	DL: LSTM; CNN; Traditional Neural Network Trans: HingMBERT	Yes	Metrics: MCC; weighted-F1; macro-F1; Accuracy Macro-F1: Yes Acc: Yes	LSTM achieved 0.72 macro F1, 63.5% accuracy
Leveraging transformers for hate speech detection in conversational code-mixed tweets.	2021	HASOC 2021 Subtask 2 dataset (Train: 5,740 conversations; Test: 1,348 conversations)	Hinglish-Code-Mixed [Both]	Twitter	Hate Speech and Offensive Content Detection in Conversations [Hate Type: Both]	Soft Voting; Hard Voting	DL: Transformer Trans: Indic-BERT; XLM-RoBERTa; Multilingual BERT	No	Metrics: Macro Precision; Macro Recall; Macro F1; Accuracy Macro-F1: Yes Acc: Yes	Hard Voting Ensemble of Indic-BERT, XLM-RoBERTa, and Multilingual BERT won first place on leaderboard (F1: 0.7253).
Mind your language: Abuse and offense detection for code-switched languages.	2019	Hinglish Offensive Tweets (HOT) (Not Reported)	Hinglish-Code-Mixed [Roman]	Not Reported	Abuse and Offense Detection [Hate Type: Both]	None	DL: LSTM	No	Metrics: Accuracy	LSTM (transfer learning) achieved 87% accuracy
Multilingual offensive language identification with cross-lingual embeddings.	2020	OLID; Spanish (Hateval 2019); Bengali; Hindi (HASOC 2019) (Bengali: 4,000; English: 14,100; Hindi: 8,000; Spanish: 6,600)	English; Hinglish-Code-Mixed; Hindi; Other [Both; Other]	Facebook; Twitter	Offensive Language Identification [Hate Type: Both]	None	DL: Transformer Trans: XLM-RoBERTa; BERT; ELMO; BERT-m	Yes	Metrics: Not Reported	Evaluates cross-lingual transformer weights learned on English to initialize models for lower-resource Hindi, Spanish, Bengali.

Challenges and approaches of code-mixed hate speech detection.	2023	HASOC 2019 (English: 7,510; German: 5,463; Hindi: 6,119; Multilanguage: 19,092)	English; German; Hindi [Both; Other]	Twitter; Facebook	Multilingual Hate Speech Detection and Sentiment Analysis [Hate Type: Both]	RF; LR; NB; SVM; k-NN; J48graft	DL: Hyb-DGRNN (Hybrid diagonal gated recurrent neural network)	Yes	Metrics: Accuracy; Precision; Recall; F-measure; Specificity Macro-F1: Yes Acc: Yes	Proposed FE-DGRNN achieved accuracy of 87.74% on Task-1, 88.98% on Task-2, and 84.74% on Task-3.
NITP-AI-NLP@ HASOC-FIRE2020: Fine tuned BERT for the hate speech and offensive content identification from social media.	2020	HASOC 2020 (English (3708/814); Hindi (2963/663); German (2373/526))	English; Hindi; German [Both; Other]	Twitter; Facebook	Hate Speech and Offensive Content Identification [Hate Type: Both]	None	DL: Transformer Trans: BERT; Fine-tuned BERT	No	Metrics: F1-score; Precision; Recall Macro-F1: Yes	Fine-tuned BERT achieved test F1-scores of 0.5031 (English subtask A), 0.1623 (English subtask B), 0.5300 (Hindi subtask A).
NSIT & IIITDWD@ HASOC 2020: Deep learning model for hate-speech identification in indo-european languages.	2020	HASOC 2020 (English (3708/814); German (2373/526); Hindi (2963/663))	English; Hindi; German [Both; Other]	Twitter	Hate Speech and Offensive Content Identification [Hate Type: Both]	None	DL: CNN; BiLSTM; CNN+BiLSTM (hybrid)	No	Metrics: f1 macro-avg score Macro-F1: Yes	Secured First and Second positions for Hindi sub-tasks A and B (F1 macro-avg scores of 0.5337 & 0.2667).
On importance of code-mixed embeddings for hate speech identification	2024	HASOC 2021 Hindi-English; Hate dataset (Not Reported)	Hinglish-Code-Mixed [Roman]	Twitter	Hate Speech and Offensive Content Detection [Hate Type: Both]	Random Forest; Linear Regression; SVM; KNN	DL: Transformer; HingFT Trans: BERT; HingBERT; Hing-mBERT; Hing-RoBERTa; HingGPT;	Yes	Metrics: F1-score; Precision; Accuracy; Recall Macro-F1: Yes Acc: Yes	HingRoBERTa pre-trained on Romanized Hinglish L3Cube-HingCorpus outperformed standard BERT (F1:
Online multilingual hate speech detection: Experimenting with hindi and english social media.	2021	homogeneous dataset (76,403 samples)	English; Hindi; Hinglish-Code-Mixed [Both]	Twitter	Multiclass Hate Speech Classification [Hate Type: Both]	Logistic Regression; SVM; Random Forest	DL: CNN; BiLSTM Trans: BERT; RoBERTa; XLM-R	Yes	Metrics: F1-score; Accuracy; Precision; Recall Macro-F1: Yes Acc: Yes	BERT achieved accuracies of 80% on EN, 95% on HI, and 86% on Hinglish.
QutNocturnal@ HASOC'19: CNN for Hate Speech and Offensive Content Identification in Hindi Language	2020	HASOC 2019 Hindi (5,984 samples)	Hindi; Hinglish-Code-Mixed [Both]	Twitter	Binary Hate Speech Classification [Hate Type: Both]	SVM; RF; XGB; MNB; KNN; RC	DL: CNN; LSTM; DNN	No	Metrics: F1-score; Precision; Recall; Accuracy Macro-F1: Yes Acc: Yes	Custom CNN with task-specific word2vec achieved 0.81 F1 and 0.82 Acc on test set.
Revisiting hate speech benchmarks: From data curation to system deployment.	2023	GOTHate (51,367 tweets)	English; Hindi; Hinglish-Code-Mixed [Both]	Twitter	Multiclass Hate Speech Classification [Hate Type: Both]	NB; LR; SVC; Davidson model	DL: CNN; LSTM; RNN Trans: mBERT; ARHNet; HurtBERT; SBERT; HEN-mBERT	Yes	Metrics: Macro F1; Precision; Recall; Accuracy; MCC; ROC-AUC Macro-F1: Yes Acc: Yes	HEN-mBERT outperforms baselines by 2.5% macro F1 and 5% on hate class F1.
TABHATE: A target-based hate speech detection dataset in hindi.	2024	TABHATE (2,020 tweets)	Hindi [Devanagari]	X (Twitter)	Multiclass Target-Based Hate Speech Detection [Hate Type: Both]	None	DL: CNN; LSTM; Bi-LSTM Trans: mBERT; MuRIL; IndicBERT; XLM-RoBERTa	No	Metrics: F1-score; Precision; Recall Macro-F1: Yes	XLM-RoBERTa achieved macro F1 of 0.77 and weighted F1 of 0.91 post-SMOTE.
Thar-targeted hate speech against religion: A high-quality hindi-english code-mixed dataset with the application of deep learning models for automatic detection.	2024	THAR (11,549 comments)	Hinglish-Code-Mixed [Both]	YouTube	Binary & Multiclass Targeted Religious Hate Speech [Hate Type: Both]	None	DL: CNN; LSTM Trans: mBERT; IndicBERT; MuRIL	No	Metrics: F1-score; Precision; Recall Macro-F1: Yes	MuRIL achieved macro F1 of 0.78 for Subtask-1 and 0.65 for Subtask-2.

The impact of data pre-processing on hate speech detection in a mix of english and hindi-english (code-mixed) tweets.	2023	Combined dataset (32,551 tweets)	English; Hinglish-Code-Mixed [Roman]	Twitter	Binary Hate Speech Detection [Hate Type: Both]	None	DL: CNN; LSTM (CNN-LSTM)	No	Metrics: Accuracy; Test Loss; Runtime; Precision; Recall; F1-score Macro-F1: Yes Acc: Yes	CNN-LSTM with over-sampling achieved 88.97% accuracy and halved the runtime.
Towards safer online spaces: Deep learning for hate speech detection in code-mixed social media conversations.	2024	HASOC 2023 Task 2 (13,996 train / 998 test)	Hinglish-Code-Mixed [Both]	Twitter	Binary & Multiclass Conversational Hate Detection [Hate Type: Both]	None	DL: LSTM; BiLSTM Trans: mBERT; SentBERT; MuRIL	No	Metrics: Macro F1; Recall; Precision; Validation Accuracy Macro-F1: Yes Acc: Yes	mBERT-BiLSTM ensemble model achieved F1 scores of 0.7008 (2a) and 0.5631 (2b).
Uncovering political hate speech during indian election campaign: A new low-resource dataset and baselines.	2022	IEHate (11,457 tweets)	Hindi [Devanagari]	Twitter	Binary Political Hate Speech Detection [Hate Type: Both]	NB; Decision Tree; Logistic Regression; SVM	DL: LSTM; BiLSTM; BiLSTM+CNN Trans: XLM-RoBERTa-base; BERT (HAM, HCMAM); RoBERTa	No	Metrics: Accuracy; Macro F1-score; MMAE Macro-F1: Yes Acc: Yes	RoBERTa achieved best performance with F1 score of 0.725 and accuracy of 0.923.
User-aware multilingual abusive content detection in social media.	2023	SCIDN; MACI (1.5M / 665K comments)	English; Hindi; Hinglish-Code-Mixed; Other [Both]	ShareChat; Moj	Binary User-Aware Abusive Content Detection [Hate Type: Both]	SVM; LR; Random Forest	DL: MLP; LSTM Trans: MuRIL; mBERT; XLM-R	No	Metrics: Accuracy; Precision; Recall; F1-score Macro-F1: Yes Acc: Yes	Joint Feature Learning fusing text & social context got F1 of 95.38% (SCIDN) and 91.58% (MACI).
Overview of the HASOC subtrack at FIRE 2021: Hate speech and offensive content identification in english and indo-aryan languages and conversational hate speech.	2021	HASOC 2021 (English, Hindi, Marathi, ICHCL)	English; Hindi; Hinglish-Code-Mixed; Other (Marathi) [Both]	Twitter	Binary & Multiclass Conversational Hate Speech [Hate Type: Both]	SVM with TF-IDF	DL: CNN; LSTM; BiLSTM; ULMFit Trans: Indic-BERT; XLM-RoBERTa; mBERT; BERT; DistilBERT; RoBERTa; ALBERT	Not Reported	Metrics: Macro F1; Accuracy; Precision; Recall Macro-F1: Yes Acc: Yes	Overview of track; best systems used transformer ensembles (Marathi F1: 0.9144).
Overview of the hasoc subtrack at fire 2022: Hate speech and offensive content identification in english and indo-aryan languages.	2022	HASOC 2022; MOLD 2.0 (Codemix; German; Marathi)	Hinglish-Code-Mixed; Other (German, Marathi) [Both]	Twitter	Binary & Multiclass Conversational Hate Speech [Hate Type: Both]	Random Forest; Logistic Regression	Trans: Google-MuRIL; XLM-RoBERTa; Indic-BERT; BERT	Not Reported	Metrics: Macro F1; Precision; Recall Macro-F1: Yes	XLM-RoBERTa base model performed best on conversational, RF on small Marathi.
Detecting hate speech for Hindi-English code-mix text data using dual contrastive learning.	2025	TRAC-1 Dataset (8,626 Romanized Hi-En comments (from 12,000 FB/Twitter posts))	Hindi-English Code-Mixed [Romanized / Latin]	Facebook; Twitter	Binary Hate Speech Classification [Hate Type: Both (Overt and Covert Aggression)]	KNN; Decision Tree (DT); Logistic Regression (LR)	DL: Dual Contrastive Learning (DCL) with Self-Supervised & Supervised stages Trans: SentBERT (Sentence-BERT)	No (In-domain balanced TRAC-1 split)	Metrics: Accuracy; Precision; Recall; F1-Score Macro-F1: Yes (F1: 91%) Acc: Yes (Acc: 86.00%)	SentBERT + Dual Contrastive Learning (DCL) achieved 86.00% accuracy and 91% F1-score, outperforming traditional ML baselines.

byteSizedLLM@NLU of Devanagari Script Languages 2025: Hate speech detection and target identification using customized Attention BiLSTM and XLM-RoBERTa base embeddings.	2025	CHiPSAL 2025 Subtasks B & C + Fine-tuning corpora (Bhojpuri, Nepali, Sanskrit, Hindi, Marathi) (Subtask B: 27,171 posts; Subtask C: 3,163 posts; Fine-tuning MLM: ~209 MB corpus)	Multilingual Devanagari (Hindi and Nepali) [Devanagari]	Social Media (Twitter/X, News, Forums)	Subtask B: Binary Hate Detection; Subtask C: 3-Class Target Identification [Hate Type: Both]	XGBoost; Random Forest; Logistic Regression; Gradient Boosting; SVC; AdaBoost; Extra Trees; Ridge; SGD; Ensemble	DL: 2-layer BiLSTM with Attention, Layer Normalization, and Dropout Trans: XLM-RoBERTa base (MLM fine-tuned across 5 Devanagari languages)	No (Standard CHiPSAL benchmark split)	Metrics: Accuracy; Macro F1-Score Macro-F1: Yes (Task B: 0.7481, Task C: 0.6715) Acc: Yes	Attention BiLSTM + XLM-RoBERTa base achieved 0.7481 Macro F1 on Task B (7th place) and 0.6715 Macro F1 on Task C (7th place), outperforming standalone XLM-RoBERTa and ML classifiers.
DII5143A@NLU of Devanagari Script Languages 2025: Detection of hate speech and targets using hierarchical attention network.	2025	CHiPSAL 2025 Subtask B & Subtask C (Subtask B: 27,171 posts; Subtask C: 3,163 posts)	Multilingual Devanagari (Hindi and Nepali) [Devanagari]	Social Media (Twitter/X, News, Forums)	Subtask B: Binary Hate Detection; Subtask C: 3-Class Target Identification [Hate Type: Both]	None	DL: Word-level & Sentence-level Dual Attention + Hierarchical Gated Adaptive Weighting (HGAA) + MLP classification head	No (Standard CHiPSAL benchmark split)	Metrics: Accuracy; Precision; Recall; Macro F1-Score Macro-F1: Yes (Subtask B: 0.745, Subtask C: 0.658) Acc: Yes (Subtask	Hierarchical Gated Adaptive Weighting XLM-RoBERTa achieved 0.745 F1 / 0.890 Accuracy on Subtask B (Rank 8) and 0.658 F1 /
IITR-CIOL@NLU of Devanagari Script Languages 2025: Multilingual hate speech detection and target identification in Devanagari-scripted languages.	2025	CHiPSAL 2025 Subtask B & Subtask C (Subtask B: 27,171 posts; Subtask C: 3,163 posts)	Multilingual Devanagari & Transliterated (Hindi, Marathi, Nepali, Bhojpuri, Sanskrit) [Devanagari and Transliterated Roman]	Social Media (Twitter/X)	Subtask B: Binary Hate Detection; Subtask C: 3-Class Target Identification [Hate Type: Both]	None	DL: Deep Neural Network with MLP Pre-classifier, ReLU, Dropout (0.3), and Linear Classifier Trans: ia-multilingual-transliterated-roberta (IBM pretrained)	No (Standard CHiPSAL benchmark split)	Metrics: Accuracy; Weighted/Micro Precision; Recall; F1-Score Macro-F1: Yes (Subtask B: 0.6762, Subtask C: 0.5816) Acc: Yes (Subtask B: 0.8840, Subtask C: 0.6611)	MultilingualRoberta Class achieved 88.40% Accuracy / 0.6762 F1 on Subtask B test set and 66.11% Accuracy / 0.5816 F1 on Subtask C test set.
LLMsAgainstHate@NLU of Devanagari Script Languages 2025: Hate speech detection and target identification in Devanagari languages via parameter efficient fine-tuning of LLMs.	2025	CHiPSAL 2025 Subtask B & Subtask C (Subtask B: 27,171 posts; Subtask C: 3,163 posts)	Multilingual Devanagari (Hindi and Nepali) [Devanagari]	Social Media, News, Forums	Subtask B: Binary Hate Detection; Subtask C: 3-Class Target Identification [Hate Type: Both]	None	DL: Parameter-Efficient Fine-Tuning (PEFT) via Low-Rank Adaptation (LoRA rank=16, alpha=16) with 4-bit quantization Trans: Mistral Nemo-Instruct-2407 (6.97B); Llama-3.1-8B; Qwen2.5-7B-Instruct; Phi3-	No (Standard CHiPSAL benchmark split)	Metrics: Accuracy; Weighted Average F1-Score Macro-F1: Yes (Weighted F1: Task B 90.05%, Task C 71.47%) Acc: Yes (Task B: 90.75%, Task C: 72.00%)	Quantized Mistral Nemo-Instruct-2407 fine-tuned via LoRA achieved top LLM performance: 90.75% Accuracy / 90.05% Weighted F1 on Task B and 72.00% Accuracy / 71.47% Weighted F1 on Task C.
MDSBots@NLU of Devanagari Script Languages 2025: Detection of language, hate speech, and targets using MURTweet.	2025	CHiPSAL 2025 Subtasks A, B, C (+ Synthetic GPT-4 Community Augmentation) (Task A: 52,422; Task B: 27,171; Task C: 3,163 (+ 330 synthetic	Multilingual Devanagari (Nepali, Marathi, Sanskrit, Bhojpuri, Hindi) [Devanagari]	Twitter (X)	Task A: 5-Class Language ID; Task B: Binary Hate; Task C: 3-Class Target ID [Hate Type: Both]	TF-IDF + Logistic Regression; SVM; Random Forest; XGBoost	DL: Rule-based Named Entity Recognition (NER) tagger + Fine-tuned Transformer Trans: MuRIL Large (24-layer, MURTweet); MuRIL Base; XLM-	Yes (Synthetic LLM augmentation + Entity tagging)	Metrics: Precision; Recall; F1-score; Accuracy Macro-F1: Yes (Task A: 0.9968, Task B: 0.7662, Task C: 0.7098) Acc: Yes (Task A: 0.9972, Task B:	MURTweet / NERMURTweet won 1st Place in Subtask C (0.7098 Macro F1 / 76.84% Acc), 3rd Place in Subtask B (0.7662 Macro F1 / 90.28% Acc), and 6th Place
Nepali Transformers@NLU of Devanagari Script Languages 2025: Detection of language, hate speech and targets.	2025	CHiPSAL 2025 Tasks A, B, C (+ Aya Expanse 8B LLM Synthetic Augmentation) (Task A: 52,422; Task B: 27,171	Multilingual Devanagari (Hindi, Nepali, Marathi, Sanskrit, Bhojpuri) [Devanagari]	Twitter (X)	Task A: Language ID; Task B: Hate Detection; Task C: Target ID [Hate Type: Both]	TF-IDF / GloVe / Word2Vec + Logistic Regression; Random Forest; Decision Tree; SVM; XGBoost; AdaBoost	DL: Bayesian Optimized Fine-tuned Transformers Trans: Twitter XLM-RoBERTa (TweetEval); BERT-base; RoBERTa;	Yes (Aya Expanse 8B synthetic generation)	Metrics: Accuracy; F1-Score Macro-F1: Yes (Task A: 99.5%, Task B: 72.5%, Task C: 61.8%) Acc: Yes (Task A:	Twitter-trained XLM-RoBERTa achieved 99.5% F1 on Task A, 72.5% F1 on Task B (Rank 14), and 61.8% F1 on Task C (Rank

NLPineers@NLU of Devanagari Script Languages 2025: Hate speech detection using ensembling of BERT-based models.	2025	CHiPSAL 2025 Subtask B (+ Backtranslated mBART-large-50 Augmentation) (19,019 Train + 4,076 Eval (Augmented to 13,695 training instances))	Multilingual Devanagari (Hindi and Nepali) [Devanagari]	Twitter (X)	Subtask B: Binary Hate Speech Detection [Hate Type: Both]	FastText + Logistic Regression; XLM-RoBERTa + Logistic Regression	DL: Hierarchical Fallback Ensemble (Primary: XLM-R, Secondary: MuRIL, Fallback: MuRIL-abusive); IndicBERT + LSTM-CNN; FastText + LSTM Trans: XLM-	Yes (mBART backtranslation + Cosine Similarity filtering)	Metrics: Recall; Precision; F1-Score; Accuracy Macro-F1: Yes (Ensemble F1: 0.6914, Test Recall: 0.7762) Acc: Yes (Ensemble Acc: 82.58%)	Hierarchical Fallback Ensemble (M7 XLM-R + M3 MuRIL + M1 MuRIL-abusive) achieved top Recall of 0.7762 (Rank 3/31) and 0.6914 Macro F1 (Rank 17/31).
Paramananda@NLU of Devanagari Script Languages 2025: Detection of language, hate speech and targets using FastText and BERT.	2025	CHiPSAL 2025 Benchmark Dataset (Subtasks A, B, C) (Task A: 52,422; Task B: 19,019 (Augmented with Task C); Task C: 2,214)	Multilingual Devanagari (Nepali, Marathi, Sanskrit, Bhojpuri, Hindi) [Devanagari (Extracted raw script)]	Social Media (Twitter/X)	Task A: Language ID; Task B: Binary Hate; Task C: 3-Class Target ID [Hate Type: Both]	FastText (100-dim, subword n-gram, word n-gram=1, bucket=10,000)	DL: Fine-tuned Transformer Trans: BERT Base Multilingual Uncased; BERT Base Uncased	Yes (Merged Subtask C into Subtask B for data augmentation)	Metrics: F1 Score; Accuracy; Precision; Recall Macro-F1: Yes (Task A: 0.9917, Task B: 0.8552, Task C: 0.5785) Acc: Yes (Task B)	FastText + Data Augmentation won 1st Place on Subtask B with 0.8552 Test F1 (vs. BERT's 0.5763) and 0.9917 F1 on Subtask A. BERT
DweshVaani: An LLM for detecting religious hate speech in code-mixed Hindi-English.	2025	Targeted Hate Speech Against Religion (THAR) + HateCheckHIn Suite (11,549 YouTube comments (10,394 Train / 1,155 Test))	Hindi-English Code-Mixed (Hinglish) [Both (Devanagari and Romanized Latin)]	YouTube	Subtask 2: 4-Class Targeted Religion Identification (Islam, Hinduism, Christianity, None) [Hate Type: Both (Targeted Religious Hate)]	None	DL: QLoRA (4-bit quantization, rank 64, alpha 128) fine-tuning; RAG with MuRIL-large-cased sentence embeddings (K=1, 4, 8); Knowledge	Yes (Black-box testing on HateCheckHIn across 35 functional tests)	Metrics: Micro F1 Score; Macro F1 Score; Functional Accuracy Macro-F1: Yes (DweshVaani Macro F1: 64.27%, Micro F1: 76.19%)	DweshVaani (Gemma-2 9B QLoRA) achieved state-of-the-art 76.19% Micro F1 / 64.27% Macro F1, outperforming fine-tuned MuRIL
Natural language understanding of Devanagari script languages: Language identification, hate speech and its target detection.	2025	CHiPSAL 2025 Benchmark Dataset (Subtasks A, B, C) (Subtask A: 74,889; Subtask B: 27,171; Subtask C: 3,163)	Multilingual Devanagari (Nepali, Marathi, Sanskrit, Bhojpuri, Hindi) [Devanagari]	Twitter/X, Facebook, News, Forums	Subtask A: Language ID; Subtask B: Hate Detection; Subtask C: Target ID [Hate Type: Both]	FastText (CUFE); TF-IDF + Classical ML (LR, SVM, RF, MNB)	DL: BiLSTM + Attention; Hybrid NER + Transformer; CNN; Ensemble Stacked Trans: mBERT; XLM-RoBERTa; IndicBERT; IndicBERT V2;	Yes (Multi-corpus benchmark compilation)	Metrics: Accuracy; Precision; Recall; Macro F1-Score Macro-F1: Yes (Subtask A: 99.97%, Subtask B: 91.36% / 81.39% P, Subtask C: 76.84% / 70.98% F1)	Shared task overview summarizing 113 registered participants and 32 submitting teams. Top winners: CUFE (FastText 99.97% F1 in Task A),
Prototypical network based few shot learning to detect Hindi-English code-mixed offensive text.	2025	HBN-tweet Dataset; TRAC-2 2020; HASOC Hate Speech Dataset (HBN-tweet: 660 Hi-En tweets; TRAC-2: 5,184 posts; HASOC: 4,579 posts)	Hindi-English Code-Mixed (Hi-En) [Romanized / Latin script mixed with Devanagari]	Twitter (X)	Tri-class Classification (Hate, Bodyshame, None) & Aggression Identification [Hate Type: Both (Hate Speech and Body Shaming)]	Logistic Regression; Multinomial Naive Bayes	DL: Weakly Supervised Few-Shot Learning with Episodic Training + MuRIL BERT + 1D CNN + Global Max Pooling + Cosine Similarity Siamese Prototypical	Yes (Evaluated across HBN-tweet, TRAC-2, and HASOC datasets)	Metrics: Accuracy; Cosine Similarity; Loss Acc: Yes (80-shot Acc: HBN 72.66%, Hate Speech 79.50%, TRAC-2 73.33%)	CNN-Siamese-FSL outperformed Machine Learning baselines and Matching Networks significantly across 10-shot, 50-shot, and 80-shot weak supervision, proving
Fighting Hate Speech from Bilingual Hinglish Speaker's Perspective, a Transformer-and Translation-Based Leveraging weakly annotated data for hate speech detection in code-mixed Hinglish: A feasibility-driven transfer learning approach with large	2022[1]	Bohra et al. (2018) dataset; 4,575 code-mixed tweets (1,661 Hate speech, 2,914 Non-Hate speech)[2]; 70:30 train-test split[3]	Hindi-English Code-Mixed (Hinglish)[1]; Roman script translated and transliterated into Devanagari script[4]	Twitter[2][6]	Hate Speech Identification / Binary Classification (Hate vs. Non-hate) [1][7]	Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), Naïve Bayes (NB), K Nearest Neighbors	DL: Deep Neural Network (DNN)[1][8] Trans: Proposed TIF-DNN[1][9]; mBERT, IndicBERT, pre-trained BERT	No[2][6]	Metrics: Accuracy[1][3]Accuracy: 73% (TIF-DNN)[1][12], 72% (SVM on translated data)[13], 71% (pre-trained BERT)[3]. 67%	Proposed TIF-DNN achieved state-of-the-art accuracy of 73%[1][12], outperforming CNN-LSTM (72%), Subword LSTM (71%).
	2024[14][15]	YouTube comments dataset[14]; 100 comments (Weakly annotated; Coarse: 25 Misogynistic 'MGY', 75 Not misogynistic 'NOT'[17]; Fine-	English & mix-code Hinglish[14][17]; Roman script[15]	YouTube[14][16]	Coarse-grained (binary: misogynistic vs. not) & fine-grained multi-label misogyny classification[14][17]	None[14][17]	DL: Transformer & LLMs[14][15]Trans / LLMs: BART large (zero-shot NLI framework)[14][19], ChatGPT-3 / GPT-3 (zero-shot, one-shot, few-shot	No[16][17]	Metrics: Accuracy, Macro-F1, Precision, Recall, MCC[21][22]Coarse Scores: • Zero-shot BART: Accuracy = 0.54, Macro-F1 = 0.5245[22]• Few-	Zero-shot classification using BART large and few-shot/one-shot prompting via ChatGPT-3 performed best among transfer

Overview of the HASOC Track at FIRE 2019: Hate Speech and Offensive Content Identification in Indo-European Languages	2019[27]	HASOC 2019 dataset[27](DE: Train/Dev 3,819, Test 850; EN: Train/Dev 5,852, Test 1,153; HI: Train/Dev 4,665, Test 1,318)[27]	English; German; Hindi[27]	Twitter; Facebook [27]	Hate Speech & Offensive Content Identification[27] (Sub-task A: HOF vs NOT; Sub-task B: HATE/OFFN/PRFN; Sub-task C: TIN vs UNTV[27])	Various ML and ensemble baselines used by participating teams [30]	DL: CNN, LSTM[30] Trans: BERT[32]	No[27]	Metrics: Macro-F1, Weighted F1[29][33] Best Macro-F1 Scores:• EN Sub-task A: 0.7882[30]• DE Sub-task A: 0.8149[31]• HI Sub-task A: 0.6162[32]	Established the HASOC 2019 shared task benchmark for evaluating hate speech and offensive content identification across English, German
Detecting Hate Speech in Hindi in Online Social Media	2023[1]	HateCheckHIn (HASOC-2021 & CONSTRAINT-2021 combined; 5,884 tweets)	English; Hindi; Hinglish-Code-Mixed [Script: Both]	Twitter	Hate Speech Detection (Normal speech vs. Hate speech)	Support Vector Machine (SVM); Naive Bayes; Decision Tree; Random Forest[41]	DL: None / NR Trans: None / NR	Yes	Metrics: Accuracy; Precision; Recall; F1-Score Accuracy: 96%[41]	SVM achieved 96% accuracy using text features (length, user mentions, punctuation, and
mBERT-GRU multilingual deep learning framework for hate speech detection in social media	2023	HS (Bohra et al., 2018); HASOC 2019 (Mandl et al., 2019); HEOT (Mathur, Shah, et al., 2018)	Hinglish / Hindi-English code-mixed	Social media / Twitter-based datasets	Hate Speech Detection	none	mBERT contextual embeddings + BiGRU	yes	Macro-F1: 0.77; Weighted-F1: 0.78	mBERT-GRU achieved macro F1 of 0.77 and weighted F1 of 0.78; the BiGRU component adds sequential